Functional Annotation of *ispE* (CMK / IspE; UniProt Q88PX5; PP_0723) in *Pseudomonas putida* KT2440

1. Summary / Answer to the Research Question

The *ispE* gene (locus **PP_0723**; UniProt **Q88PX5**) of *Pseudomonas putida* KT2440 encodes **4-diphosphocytidyl-2-C-methyl-D-erythritol kinase** (abbreviated **CMK** or **IspE**; **EC 2.7.1.148**). It is a cytoplasmic, ATP-dependent small-molecule kinase of the **GHMP kinase superfamily (IspE subfamily)** that catalyzes the **fourth of the eight steps** of the **2-C-methyl-D-erythritol-4-phosphate (MEP) / non-mevalonate / DOXP pathway** of isoprenoid precursor biosynthesis. Specifically, it phosphorylates the **2-hydroxyl group** of 4-diphosphocytidyl-2-C-methyl-D-erythritol (CDP-ME), using ATP (and Mg^{2+}), to give 4-diphosphocytidyl-2-C-methyl-D-erythritol 2-phosphate (CDP-ME2P) plus ADP. This committed step channels carbon toward the universal isoprenoid building blocks isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP), which in *P. putida* are used for respiratory quinones (ubiquinone/menaquinone), cell-envelope carrier lipids, carotenoids, and other terpenoids. Because the MEP pathway is essential in most bacteria yet absent from humans, IspE is an established antibacterial drug target.

Gene-identity verification (mandatory): The gene symbol *ispE*, the protein name CMK, EC 2.7.1.148, the GHMP-kinase/IspE-subfamily assignment, and the InterPro domains (IPR004424 IspE; IPR006204/IPR013750/IPR036554 GHMP N/C domains) in the UniProt record are all mutually consistent and are corroborated by the primary literature on IspE from *E. coli*, *Plasmodium*, and plants. No conflicting "same-symbol, different-gene" literature was encountered. There is no *P. putida* KT2440–specific experimental paper on this enzyme; the functional assignment for Q88PX5 rests on the HAMAP rule MF_00061 and on high orthology to biochemically and structurally characterized IspE enzymes (chiefly *E. coli*), which is the standard and robust basis for annotating this highly conserved housekeeping enzyme.

2. Primary Function: The Catalyzed Reaction and Substrate Specificity

Reaction (EC 2.7.1.148):

> 4-diphosphocytidyl-2-C-methyl-D-erythritol (CDP-ME) + ATP → 4-diphosphocytidyl-2-C-methyl-D-erythritol 2-phosphate (CDP-ME2P) + ADP

The reaction was first demonstrated biochemically by Lüttgen *et al.* (2000), who identified the then-unannotated *E. coli ychB* gene (= *ispE*) as a MEP-pathway enzyme. Comparative genomics showed that *ychB* orthologs co-distribute with *dxs*, *dxr*, and *ygbP* (*ispD*) — genes already known to encode enzymes of the deoxyxylulose-phosphate pathway — suggesting *ychB* also belongs to that pathway. The recombinant protein, purified to homogeneity, "was shown to phosphorylate 4-diphosphocytidyl-2C-methyl-D-erythritol in an ATP-dependent reaction. The reaction product was identified as 4-diphosphocytidyl-2C-methyl-D-erythritol 2-phosphate by NMR" using multiple ^{13}C-labeled substrates [PMID 10655484]. The paper's title states the regiochemistry explicitly — the enzyme "phosphorylates the 2-hydroxy group" of CDP-ME [PMID 10655484].

Substrate specificity. The physiological phosphoryl acceptor is the cytidine-containing intermediate **CDP-ME**, and the phosphoryl donor is **ATP** (Mg^{2+}-dependent, as for GHMP kinases generally). Phosphorylation is strictly **regiospecific for the 2-hydroxyl** of the methylerythritol moiety [PMID 10655484]. This narrow specificity places IspE as a dedicated, committed step of the MEP pathway rather than a promiscuous kinase.

Downstream fate confirms on-pathway role. A ^{14}C-labeled sample of the IspE product (CDP-ME2P) "was converted efficiently into carotenoids by isolated chromoplasts" of *Capsicum annuum* [PMID 10655484], demonstrating that the product is a genuine on-pathway isoprenoid intermediate. In the pathway, CDP-ME2P is next cyclized by IspF (2-C-methyl-D-erythritol 2,4-cyclodiphosphate synthase) with release of CMP [PMID 29335298], then reduced by IspG and IspH to yield IPP and DMAPP.

3. Structure, Family, and Catalytic Mechanism

Family and fold. IspE belongs to the **GHMP kinase superfamily** (named for Galactokinase, Homoserine kinase, Mevalonate kinase, and Phosphomevalonate kinase), consistent with the UniProt/InterPro domain set (GHMP N-domain IPR006204; GHMP C-domain IPR013750/IPR036554; IspE-specific IPR004424). GHMP kinases are two-lobed (N- and C-terminal domains) enzymes whose active site lies in the cleft between the domains [PMID 11751891].

Experimental structures. *E. coli* IspE has been crystallized as an **IspE–ADP binary complex** and a **ternary complex at ~2 Å resolution**; these confirm that "IspE produces 4-diphosphocytidyl-2C-methyl-D-erythritol 2-phosphate by ATP-dependent phosphorylation of 4-diphosphocytidyl-2C-methyl-D-erythritol" and place the ATP and CDP-ME substrates in the bilobed cleft [PMID 20208151]. Reassessment of the quaternary structure showed that an apparent "triclinic dimer" is a crystal-lattice artefact whose interface occludes the substrate sites, supporting a **functional monomer** [PMID 20208151]. Structural modeling of *Plasmodium* IspE orthologues likewise describes "a critically conserved canonical … (GHMP) domain within the active site lying in a deep cleft sandwiched between ATP and CDPME-binding domains" [PMID 29550587].

Target-specific sequence confirmation. The actual *P. putida* KT2440 protein (UniProt `Q88PX5`, `ISPE_PSEPK`, 286 aa, gene *ispE*) contains the diagnostic **GHMP-kinase glycine-rich phosphate-binding motif** `GGGIGGGSSD` (matching the GGG-x-GGGS "motif 2" consensus) in its N-terminal domain, within the context `WLTKVLPM-GGGIGGGSSD-AATTL`. This is the loop that cradles the ATP phosphates for phosphoryl transfer in GHMP kinases, directly corroborating — at the level of the target sequence itself, not merely by orthology — the HAMAP (MF_00061)/InterPro assignment of Q88PX5 to the GHMP-kinase family/IspE subfamily (analysis of the UniProt sequence).

Catalytic mechanism (inferred from GHMP-family structures). The active-site cleft contains conserved motifs including a **glycine-rich phosphate-binding loop**, and an **invariant aspartate** near the phosphate-binding site serves as the catalytic residue — in *Methanococcus jannaschii* mevalonate kinase, "Asp(155), an invariant residue …, is located close to the putative phosphate-binding site and has been assumed to play the catalytic role" [PMID 11751891]. QM/MM analysis of the related phosphomevalonate kinase shows that a conserved lysine "functions to neutralize the negative charge developed at the β-, γ-bridging oxygen atom of ATP during phosphoryl transfer," reflecting "the common catalytic mechanism of the GHMP kinase superfamily" [PMID 27676321]. Applied to IspE, the model is: the catalytic aspartate deprotonates the **2-OH** of CDP-ME, generating an alkoxide that performs in-line nucleophilic attack on the ATP γ-phosphate (stabilized by Mg^{2+}, the phosphate-binding loop, and the conserved lysine), transferring the phosphate to yield CDP-ME2P + ADP. This explains the strict 2-OH regiospecificity observed experimentally [PMID 10655484].

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4. Pathway Context and Biological Processes

Pathway position. IspE is the **fourth enzyme of the MEP (non-mevalonate) pathway** [PMID 29550587]. The full route is: DXS → DXR (IspC) → IspD (CDP-ME synthase) → **IspE (CMK)** → IspF → IspG → IspH → IPP + DMAPP. IspE therefore sits at the committed midpoint, converting the cytidylylated intermediate CDP-ME to CDP-ME2P before ring closure by IspF [PMID 29335298]. The pathway "relies on eight enzyme-catalyzed stages exploiting a range of cofactors and metal ions" [PMID 21619509].

End-products and physiology. The MEP pathway supplies IPP and DMAPP, and "all isoprenoids are synthesized by the consecutive condensation of the five-carbon monomer isopentenyl diphosphate (IPP) to its isomer, dimethylallyl diphosphate (DMAPP)" [PMID 22450534]. In bacteria these building blocks are used for functions including "electron transport (e.g., ubiquinone), and cell wall biosynthesis intermediates" [PMID 22450534] (e.g., the undecaprenyl-phosphate carrier lipid for peptidoglycan/O-antigen assembly), as well as carotenoids. Direct isotope-labeling evidence in *E. coli* — a Gram-negative that, like *Pseudomonas*, uses only the MEP pathway — shows that MEP intermediates processed through IspE are incorporated into product isoprenoids: labeled 2-C-methyl-D-erythritol hydrogens appear "without any loss in the prenyl chain of menaquinone and/or ubiquinone on the carbon atoms derived from C-4 of isopentenyl diphosphate (IPP) and on the E-methyl group of dimethylallyl diphosphate (DMAPP)" [PMID 10698701]. Thus IspE activity directly underpins **respiratory quinone biosynthesis** (aerobic/anaerobic electron transport) and cell-envelope biogenesis in *P. putida*.

Essentiality. The mevalonate-independent pathway "does not occur in humans, but is present and has been shown to be essential in many dangerous pathogens" [PMID 20208151], and the fourth enzyme "4-diphosphocytidyl-2C-methyl-d-erythritol kinase (IspE) has been proved essential in pathogenic bacteria" [PMID 29550587]. Because *P. putida* is a Gram-negative Proteobacterium possessing a complete MEP pathway and lacking the mevalonate route, IspE is expected to be essential for viability by orthology.

Subcellular localization. In bacteria the MEP pathway operates in the **cytoplasm** (there is no signal/transit peptide in the IspE sequence). The distinction is illustrated by the plant/plastid situation, where the orthologue is compartmentalized: the rice OsIspE "encoded protein is targeted to the chloroplast" [PMID 29893915]. For a non-plastid-bearing Gram-negative bacterium such as *P. putida*, the enzyme carries out its function as a soluble cytoplasmic protein.

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5. Translational Significance (Drug Target)

Because the MEP pathway is essential in bacteria and absent in humans, "IspE catalyzes the ATP-dependent phosphorylation of 4-diphosphocytidyl-2-..." and its pathway is regarded as an attractive antibacterial target [PMID 41695590]. Structure-based virtual screening on the *E. coli* IspE crystal structure produced a novel inhibitor class that also showed "activity ... against the more pathogenic bacteria *Pseudomonas aeruginosa* and *Acinetobacter baumannii*" [PMID 37718320], and fragment-based campaigns continue to target IspE [PMID 41439264, 41695590]. The cross-activity against *Pseudomonas aeruginosa* supports the inference that IspE performs the same essential function in *P. putida*.

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6. Supported and Refuted Hypotheses

Supported: - **H1 — IspE is CMK (EC 2.7.1.148), the fourth MEP-pathway enzyme.** Supported by UniProt/HAMAP annotation and primary biochemistry [PMID 10655484, 29550587]. - **H2 — Substrate = CDP-ME + ATP; product = CDP-ME2P via 2-OH phosphorylation.** Supported by direct NMR product identification [PMID 10655484] and structural work [PMID 20208151]. - **H3 — GHMP two-domain fold with an inter-domain ATP/CDP-ME cleft, functional monomer.** Supported [PMID 20208151, 29550587, 11751891]. - **H4 — Cytoplasmic localization in bacteria.** Supported by orthology and the contrasting plastid targeting of the plant orthologue [PMID 29893915]. - **H5 — Product feeds IPP/DMAPP → quinones, cell-wall carrier lipids, carotenoids; pathway essential.** Supported [PMID 10698701, 22450534, 29550587, 20208151].

Refuted / set aside: - The crystallographic "dimer" as the biological unit was **refuted**; the functional enzyme is a monomer [PMID 20208151]. - No evidence supports a non-MEP or moonlighting role; IspE's specificity for CDP-ME argues against promiscuous activity.

7. Limitations and Future Directions

- **No organism-specific study:** All experimental evidence is from orthologues (*E. coli*, *Plasmodium*, plants); the *P. putida* KT2440 enzyme itself has not been individually characterized. Direct kinetic measurement (Km for CDP-ME/ATP, kcat) and gene-essentiality confirmation (e.g., transposon-insertion/conditional-knockout data) in KT2440 would strengthen the annotation.
- **Residue-level mechanism** is inferred from family homologues; an *E. coli*/*Pseudomonas* IspE ternary structure with substrate plus site-directed mutagenesis of the proposed catalytic aspartate would nail down the catalytic base.
- **PubMed search limitations:** The literature interface returned few hits for some highly specific queries; additional classic structural papers (e.g., the original *E. coli* IspE structure) likely exist beyond those retrieved here but are represented indirectly through

  P 20208151 .

8. Key References

- Lüttgen H *et al.* (2000) *PNAS* — IspE/YchB phosphorylates the 2-OH of CDP-ME.

  P 10655484
- Kalinowska-Tłuścik J *et al.* (2010) — *E. coli* IspE crystal structures; monomeric functional unit.

  P 20208151
- Kadian K *et al.* (2018) — IspE as fourth MEP enzyme, GHMP domain, ATP/CDPME cleft, essentiality.

  P 29550587
- Liu *et al.* (2018) — IspF acts on CDP-ME2P, releasing CMP (defines downstream step).

  P 29335298

- Yang D *et al.* (2002) — GHMP kinase (mevalonate kinase) fold and catalytic Asp.

  P 11751891

- Huang J *et al.* (2016) — GHMP phosphoryl-transfer mechanism (catalytic Lys).

  P 27676321

- Charon L *et al.* (2000) — MEP intermediates incorporated into ubiquinone/menaquinone.

  P 10698701

- Kuzuyama T & Seto H (2012) — isoprenoid functions; IPP/DMAPP building blocks.

  P 22450534

- Hunter WN (2011) — non-mevalonate pathway: eight steps, cofactors/metals, drug target.

  P 21619509

- Chen H *et al.* (2018) — plant OsIspE targeted to chloroplast.

  P 29893915

- Ropponen H-K *et al.* (2023) — *E. coli* IspE inhibitors active vs *P. aeruginosa*/*A. baumannii*.

  P 37718320

- Walsh *et al.* (2026) / (2025) — IspE ATP-dependent phosphorylation; MEP pathway antibacterial target.

  P 41695590

  P 41439264